EXPERIMENT – 3

Multi-Byte Data and Array Operations Using 8086 Indirect Addressing

Objective

To perform multi-byte arithmetic and array operations using 8086 indirect addressing. The experiment includes 32-bit addition, summation of N 16-bit numbers and 64-bit addition with carry propagation using SI, DI and BX registers.

Algorithm

1. Load the addresses of the operands and result using SI, DI and BX.
2. Add the lower words using ADD.
3. Add higher words using ADC to include the carry.
4. Store the result in the specified memory locations.
5. For an array, load the count into CX and process each word using LOOP.
6. For 64-bit addition, add each byte with carry propagation.
7. Verify the obtained results using hexadecimal calculations.

Software and Hardware Required

- 16-bit Emulator (EMU8086 / MASM)
- 8-bit / 16-bit Processor Trainer Kit
- Keyboard
- Power Cord

A) 32-bit Addition Using Indirect Addressing

Code

```
MOV SI,2000H
MOV DI,2004H
MOV BX,2008H
MOV AX,[SI]
ADD AX,[DI]
MOV [BX],AX
MOV AX,[SI+2]
ADC AX,[DI+2]
```

MOV [BX+2],AX

MOV AX,0000H

ADC AX,0000H

MOV [BX+4],AX

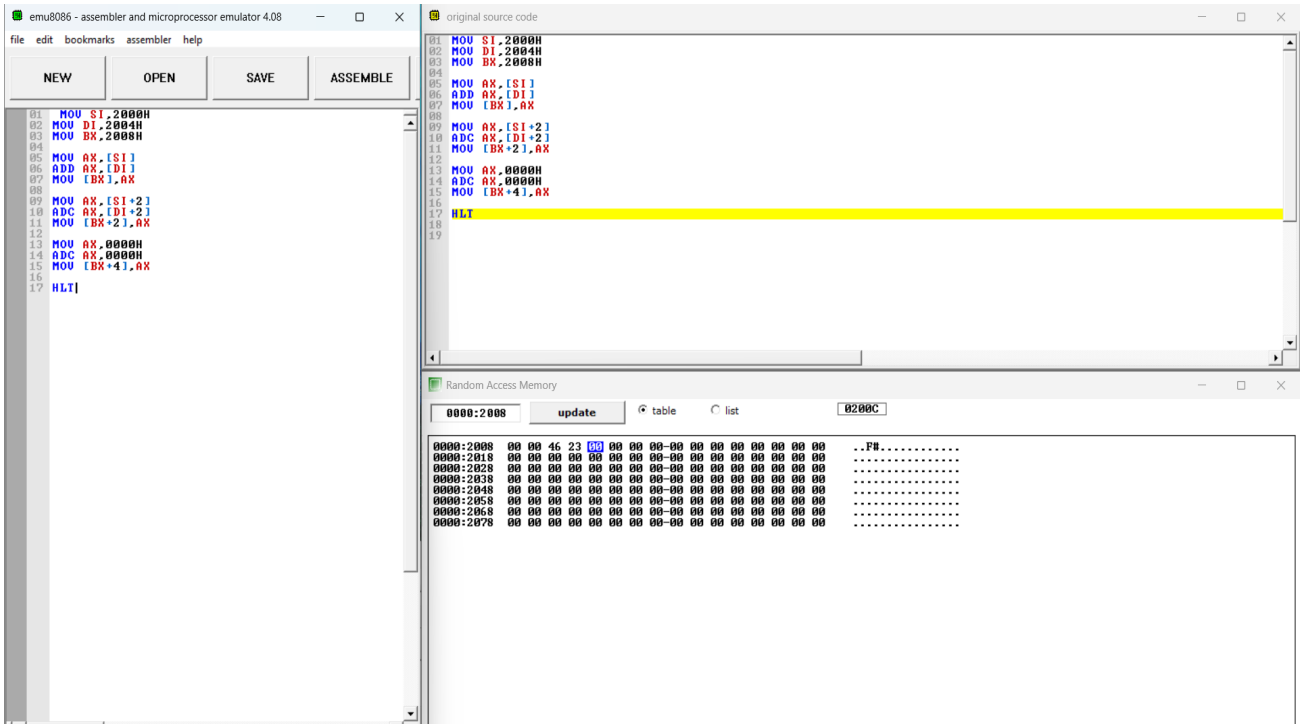
HLT

This program uses SI, DI and BX for indirect and based addressing and uses ADC for carry propagation.

INPUT

Memory Location	Data
2000H–2003H	1234 FFFFH
2004H–2007H	1111 0001H

OUTPUT



Memory Location	Data
2008H–200BH	2346 0000H
200CH	00H

Calculation:

1234 FFFFH + 1111 0001H = **2346 0000H**

B) Sum of N Numbers in an Array

Code

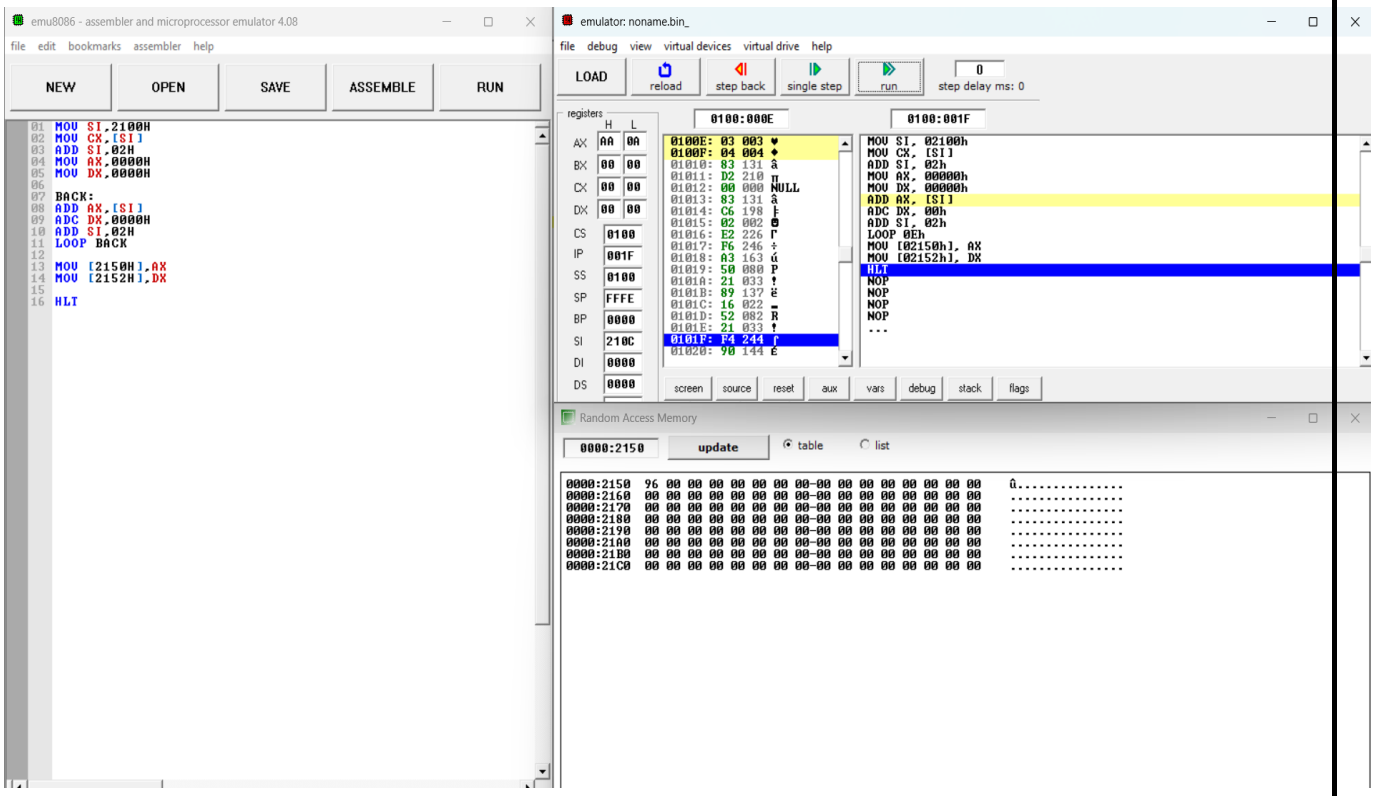
```
MOV SI,2100H
MOV CX,[SI]
ADD SI,02H
MOV AX,0000H
MOV DX,0000H
BACK:
ADD AX,[SI]
ADC DX,0000H
ADD SI,02H
LOOP BACK
MOV [2150H],AX
MOV [2152H],DX
HLT
```

The program loads N into CX, adds each 16-bit array element and uses DX:AX as a 32-bit accumulator.

Input

Memory Location	Data
2100H	0005H
2102H	000AH
2104H	0014H
2106H	001EH
2108H	0028H
210AH	0032H

Output



Memory Location	Data
2150H	0096H
2152H	0000H

Calculation:

10 + 20 + 30 + 40 + 50 = **150 = 0096H**.

C) 64-bit Multi-Byte Addition

Code

MOV SI,2200H

MOV DI,2210H

MOV BX,2220H

MOV CX,0008H

CLC

NEXT:

MOV AL,[SI]

ADC AL,[DI]

MOV [BX],AL

```
INC SI

INC DI

INC BX

LOOP NEXT

MOV AL,00H

ADC AL,00H

MOV [BX],AL

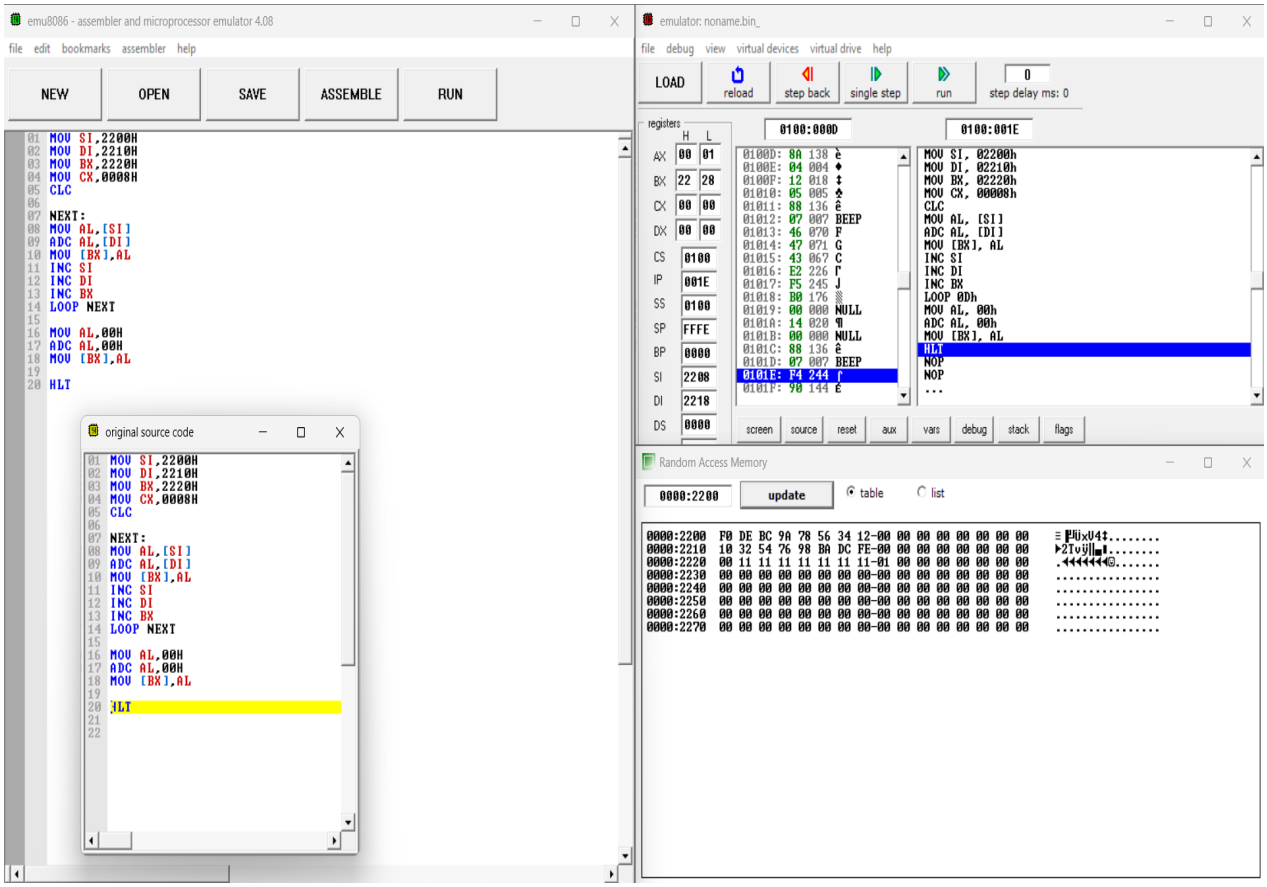
HLT
```

The program adds the two 64-bit numbers byte by byte and propagates the carry using ADC.

Input

Memory Location	Data
2200H–2207H	F0 DE BC 9A 78 56 34 12
2210H–2217H	10 32 54 76 98 BA DC FE

Output



Memory Location	Data
2220H–2227H	11 11 11 11 11 11 11 00H
2228H	01H

Calculation:

123456789ABCDEF0H + FEDCBA9876543210H
= **1111111111111100H**

Therefore:

- 64-bit result = **1111111111111100H**
- Final carry = **01H**

Result

Thus, the **32-bit addition, array summation and 64-bit multi-byte addition operations were successfully implemented using 8086 indirect addressing**. The results were stored in the specified memory locations and verified using hexadecimal calculations.

Student Reflection

I learned how to perform multi-byte arithmetic operations using the 8086 microprocessor. I understood indirect addressing using SI, DI and BX, array traversal using LOOP, and carry propagation using ADC. This experiment improved my understanding of multi-precision data processing and memory addressing.